



AL-2001 Artificial Intelligence Lab # 1

Objectives:

- Practice on Python Basics, Variables, Conditional Structure, Iterative Structure

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your logic on Copy / Notebook.
2. Write **Your Name** and **Roll No** on your Paper/Sheet's first page.
3. **Do not copy from any source otherwise you will be penalized with negative marks.**
4. Complete your lab **within given Time Slot**.
5. Paste all your codes along with screenshots in a word file and renamed with your roll number.
6. Keep all your source files in your computer for verification. Do not overwrite a single source file for all programs.

Problems:

1. Find the area of a Rectangle. (**Area = Length * Width**)
2. Write a program to calculate the percentage of student marks based on Math, Science, English, and Urdu. (**Percentage = (Obtained / Total) * 100**)
3. Calculate the Interest of a Bank Deposit (**Formula "Interest=Amount*Years*Rate/100"**)
4. Convert Temperature from Fahrenheit (**°F**) to Celsius (**°C**) (**Formula $C = 5/9 * (F - 32)$**)
5. Compute the perimeter of a rectangle. (**Perimeter = 2 * (Length * Width)**)
6. A brand offered 10% discount on each shirt purchased. The original price of the shirt is 550, find the discounted price of one shirt. Write the pseudocode for this problem statement.
7. Write a program logic that calculates the total of a retail sale. The program should ask the user for the following: the retail price of the item being purchased and the sales tax rate. Once the information has been entered the program should calculate and display the following: the sales tax for the purchase and the total sale.
8. Write a program logic that calculate the total bill of the shopping, a person purchases two keyboards each worth of 100\$, three mouse each worth of 50\$. Calculate the Total in **PKR**.



(Note: Dollar to PKR 1\$ = 303PKR)

9. Write a program that calculates the current balance in a savings account. The program should obtain from the user the following information: the starting balance, the total amount of deposits made, and the total amount of withdrawals made. After the program has calculated the current balance, it should be displayed on the screen. Assume one input for deposits and one input for withdrawals.
10. Write pseudocode to read the age of a candidate and determine whether it is eligible for casting his/her own vote or not. **Eligibility (age $\geq$ 18)**
11. Write a program to input electricity unit charges and calculate total electricity bill according to the given condition:
For first 50 units Rs. 0.50/unit
For next 100 units Rs. 0.75/unit
For next 100 units Rs. 1.20/unit
For unit above 250 Rs. 1.50/unit
An additional surcharge of 20% is added to the bill
12. Draw a flow diagram for a program that calculates the area of a shape. It asks you enter number of sides of the shape then asks you to enter the lengths. It is restricted to three major shapes. If user enters '0' as number of sides then you enter 'radius' length and program print area of the circle. If user enters '3' then program prints the area of the triangle. If user enters '4' then program prints the area of the rectangle.
13. **Write a program to check whether a given number is a perfect number or not.**
Test Data :
Input the number : 56
Expected Output :
The positive divisor : 1 2 4 7 8 14 28
The sum of the divisor is : 64
So, the number is not perfect.
14. **Write a program to find the perfect numbers within a given number of ranges.**
Test Data :
Input the starting range or number: 1
Input the ending range of number: 50
Expected Output:
The Perfect numbers within the given range: 6 28
15. **Write a program to display the n terms of square natural number and their sum. 1 4 9 16 ... n**
Terms
Test Data:



Input the number of terms: 5

Expected Output:

The square natural upto 5 terms are :1 4 9 16 25

The Sum of Square Natural Number upto 5 terms = 55

You need to done with your exercise within given time.